

DHCP Configuration — HomeSOClab

Each routed VLAN runs its own DHCP scope on pfSense (**Services → DHCP Server → [Interface]**). The Monitoring VLAN intentionally has **no DHCP** (the sensor uses static addressing; its capture NIC needs no IP at all).

Per-VLAN scopes

Interface	Subnet	Pool	Gateway	DNS	Domain
MGMT (10)	10.10.10.0/24	10.10.10.100 – .199	10.10.10.1	10.10.10.1	mgmt.soclab.local
ATTACKER (20)	10.10.20.0/24	10.10.20.100 – .199	10.10.20.1	10.10.20.1	atk.soclab.local
VICTIM (30)	10.10.30.0/24	10.10.30.100 – .199	10.10.30.1	10.10.30.1	vic.soclab.local
MONITORING (99)	10.10.99.0/24	— (disabled)	—	—	—

Settings rationale

- **Pools start at .100**: leaves .1–.99 for static infra (gateway, SIEM, analyst WS, sensor) so reserved services never collide with dynamic leases.
- **DNS = the VLAN gateway**: clients resolve via pfSense Unbound. This funnels all DNS through one observable choke point (valuable telemetry) and lets the Victim zone resolve names without any internet path.
- **Lease time**: default 7200s (2h). Short enough that lab rebuilds reclaim addresses quickly; long enough to avoid churn during exercises.
- **NTP option (042)**: hand out 10.10.X.1 so all hosts share the firewall's clock — essential for correlating timestamps across SIEM logs.

Static DHCP mappings (reservations)

Add MAC reservations for repeatable lab hosts so detections/playbooks can rely on stable IPs (**DHCP Server → [Interface] → DHCP Static Mappings**):

Host	VLAN	Reserved IP	Notes
SIEM mgmt UI	10	10.10.10.10	also set static in guest
Analyst WS	10	10.10.10.20	
Kali attacker	20	10.10.20.50	stable C2 listener IP
Victim-Win10	30	10.10.30.50	primary detonation target

The corresponding values live in `config/lab.env` . Keep them in sync.